

CEBE Interdisciplinary Fellowship Programme
Call for PhD and PostDoc Projects

Human–AI Collaborative Workflows for Immersive Digital Twins in Bridge Monitoring and Inspection (HAI-BRI)

PhD project

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PI & Main supervisor	Assoc. Prof. Giuseppe Abbiati	Department of Civil and Architectural Engineering, Aarhus University
Co-supervisor	Assist. Prof. Jhonattan Martinez	Department of Civil and Architectural Engineering, Aarhus University
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Co-supervisor	Dr. Carolin Reichherzer	Department of Civil, Environmental and Geomatic Engineering, ETH Zurich
Co-supervisor	Prof. Ueli Angst	Department of Civil, Environmental and Geomatic Engineering, ETH Zurich
Ph.D. Candidate	Lorenzo Loyola	Department of Civil and Architectural Engineering, Aarhus University

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1 Project Summary

This PhD project **HAI-BRI** will develop human–AI interaction framework for bridge Structural Health Monitoring (SHM). The framework will integrate Unmanned Aerial Systems (UAS) imagery, sensor measurements, structural information, inspection histories, environmental observations, and engineering knowledge through a Digital Twin (DT). Evidence will be semantically contextualized into an evolving representation of the bridge condition described as a Knowledge Graph (KG). Artificial Intelligence (AI) based on Large Language Models (LLM) and Vision-Language Models (VLM) will support interpretation of this evidence by identifying relevant relationships, plausible deterioration mechanisms, uncertainty, and information gaps. Extended Reality (XR) will be explored to create an immersive environment for the users.

The framework will be validated experimentally by comparing conventional engineering workflows, AI assistance without semantic grounding, and the proposed knowledge-grounded human–AI workflow. The framework will be validated on two bridge case studies in the Global South, namely, the Ponte do Funil (Itaparica Island, Bahia, Brazil), supported by the Federal University of Bahia, and the elevated viaduct sections of the Nairobi Expressway (Nairobi, Kenya), supported by the University of Nairobi. Deterioration mechanisms associated with water ingress, expansion joints, drainage deficiencies, moisture, cracking, and corrosion-related symptoms will be prioritized.

2 Project Description

2.1 Motivation, Significance and Scientific Challenges

Deterioration of bridge infrastructure is a critical issue in Western countries, where such assets, in large part, already exceed their nominal life. The catastrophic collapse of the Morandi Bridge (Genova, Italy, 2018) indicates that bridges in poor condition may still be operated. The Global South is characterized by younger infrastructure; however, underfunded maintenance programs are responsible for faster deterioration. Vibration-based SHM has been implemented extensively in the last 20 years along with the large diffusion of cost-effective sensing technology [1]. More recently, UASs have become affordable and easy to operate, facilitating the capture of high-resolution image data from hard-to-access structural elements [2]. However, engineers must still combine these heterogeneous sources to determine whether an observed condition is significant, whether additional inspection is needed, and which intervention to prioritize [3]. The overwhelming amount of data, along with the shortage of qualified personnel, renders this a challenging task.

The project’s vision is to develop a human-centered DT that supports engineers in designing and analyzing inspection campaigns. AI provides the opportunity for engineers to rapidly access and interpret a variety of data sources to support them in their decision-making process. However, effective human–AI collaboration workflows are largely underexplored for inspection and SHM workflows. Moreover, the integrating of immersive technologies to enhance user experience and decision-making for inspection workflows remains largely unexamined.

The research addresses this gap in three ways: 1) by a building a system in which AI reasoning is grounded in a semantic DT and delivered through Extended Reality (XR); 2) by developing and evaluating human–AI collaboration workflows for immersive bridge inspection; and 3) by providing an empirically grounded framework for future inspection workflows involving expected expert users.

The PhD project funded by **HAI-BRI** will address the two following research questions:

- Q1 how should the semantic representation of a bridge evolving condition be structured and integrated with AI so that reasoning on its condition is logically consistent and traceable to empirical evidence?
- Q2 how should the DT operator interact with the AI, through conversational and immersive interfaces, to build situational awareness and make better-supported inspection and maintenance decisions?

Aarhus University (AU) will tackle Q1 while ETH Zurich (ETHZ) will tackle Q2. The key-enabling technologies are 1) semantic representations of infrastructure condition, 2) multimodal LLM/VLM reasoning grounded in the DT, and 3) XR providing the immersive layer.

Q1 will be studied by comparing levels of semantic granularity and AI reasoning with and without graph-based grounding, in terms of evidence retrieval, consistency, traceability, and unsupported conclusions; Q2 by user studies and controlled experiments comparing conventional, conversational, and immersive access to the DT on decision performance, SA, cognitive workload, trust and reliance on AI.

The PhD Candidate (Lorenzo Loyola) is educated as a structural engineer and has +5 years of experience in R&D (academia and industry) related to the digitalization of structural design and monitoring workflows. He is already employed at AU. So the project will start in January 1st, 2027 and last 36m.

2.2 State of the Art

A DT combines digital and physical assets into a cyber-physical system to deliver specific services [4]. In the context of SHM, DTs are used to collect multi-modal data related to a structural system, process it, and inform decision making [5]. As of today, machine learning is used to automate some specific processing tasks (e.g., crack localization) [3]. However, decision making regarding inspection and maintenance still requires expert users to reason across large data sets, considering the entire history of the asset [6]. The SHM community recognizes that such an approach leaves substantial information unexploited and leads to sub-optimal engineering decisions. Recent work points toward the use of AI to overcome this limit [7].

Ontologies are being explored to add a semantic description to DT components and functionalities and facilitate user interaction. An ontology provides a formal, machine-interpretable representation of domain concepts and their relationships [8]. DTs, therefore, increasingly employ ontologies as semantic layers for interoperability, traceability, and machine reasoning [9, 10]. Ontologies have been recently investigated for turning fragmented inspection and maintenance data into structured knowledge for condition assessment of bridges [11, 12] with the goal of increasing Situational Awareness (SA). In fact, engineers must not only access heterogeneous information but also perceive changes in structural condition, understand their implications, and anticipate their evolution [13]. In parallel, conversational agents and ontologies are explored to operate a DT in place of a graphical user interface [12].

XR, which comprises immersive technologies such as Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), shows great promise for enhancing DTs, both as a gateway to experiencing data that simulates or extends the real world [14] and as a means of improving usability [15], collaboration, and decision-making [16]. XR is also being explored as a way to enhance the manual bridge inspection process independently of DTs [17, 18]. Immersive DT environments have been combined with interactive virtual-agent interfaces for infrastructure leak simulation and monitoring [19].

Nevertheless, important gaps remain. The outlined state of the art is fragmented, single-technology-centered, and engineers SA is overlooked [6]. Human involvement is limited to

validating AI outputs, refining generated knowledge, or facilitating interaction with digital information [7, 12], rather than being explicitly modeled as part of a joint human–AI reasoning and decision-making system.

2.3 Scientific Approach, Methodology, and Novelty

The project will develop and experimentally validate a human-centered framework for bridge SHM in which AI and XR increase operator awareness of the bridge condition. Two bridge case-studies, namely, the Ponte do Funil (Itaparica Island, Bahia, Brazil) and the elevated viaduct sections of the Nairobi Expressway (Nairobi, Kenya), are provided by the Federal University of Bahia School of Engineering (FUBSE) and the University of Nairobi (UN), respectively (Fig. 1). Both are concrete structures exposed to material deterioration (cracks and corrosion). A central research challenge will be preventing the AI from presenting unsupported conclusions

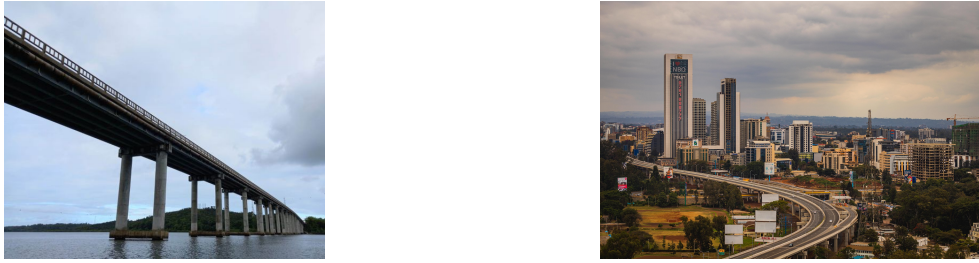


Figure 1: **HAI-BRI** case studies: Ponte do Funil, Brazil (left) and Nairobi Expressway, Kenya (right).

as established facts. A semantic description of the evolving bridge condition will constrain and contextualize AI reasoning, providing engineering-sound assessments of the bridge state and evidence for decision support. For example, when moisture is detected below an expansion joint, the AI retrieves the joint condition and inspection history, drainage configuration, recent rainfall exposure, nearby cracking or corrosion observations, and relevant engineering knowledge before proposing plausible deterioration hypotheses. In this project, the DT is an evolving representation that integrates BIM/FEM models with inspection observations, sensor measurements, environmental information, historical records, deterioration states, and maintenance information. The semantic layer that connects these heterogeneous representations is implemented as a KG. This will allow the DT to evolve as new evidence becomes available and provide a machine-interpretable representation of the infrastructure situation. The research program of **HAI-BRI** is organized into three interconnected Work Packages (WPs). **WP1 Asset knowledge representation (m1-m12)**: The main scientific contribution of WP1 will be a semantic model that links all digital assets associated with the bridge case-studies:

T1.1 Digital Twin foundation: Technical drawings, historical inspection and maintenance records, monitoring data, and other available information on the bridge case-studies will be collected and structured. A BIM model will represent geometry and structural components, while a FEM model will represent mechanical behavior. Experimental data will be organized in databases.

T1.2 Semantic layer: A semantic layer will connect the BIM/FEM models with sensors, measurements (e.g., UAS imagery, inspection records), deterioration mechanisms (e.g., cracking and corrosion), and maintenance plans. Ontologies will define the relevant concepts and relationships, which will be instantiated as a KG.

WP2 Multimodal and Graph-Grounded AI (m6-m24): The main scientific contribution of WP2 will be a multimodal, graph-grounded AI architecture for traceable infrastructure reasoning for evidence-based interpretation of bridge condition:

T2.1 Automated annotation of deterioration: WP2 will investigate how AI can transform heterogeneous observations into evidence-grounded interpretations of bridge condition. LLM/VLM will identify and contextualize candidate observations from data. The objective is not to develop another universal defect detector, but to determine how automatically extracted observations can become structured evidence within the semantic DT.

T2.2 Automated reasoning on structure condition: A graph-grounded LLM/VLM reasoning architecture will retrieve relevant asset-specific information from the KG before generating an engineering interpretation. The quality of AI-supported assessment changes as different information sources are removed or become incomplete will also be investigated.

WP3 Development and Evaluation of an Immersive Human–AI Collaboration Workflows Framework (m18-m36): WP3 will develop and evaluate methods of human–AI collaborative workflows that support trust, appropriate reliance, and decision-making for bridge inspection within immersive environments:

T3.1 Human–AI Collaboration Workflows: T3.1 develops and implements the conversational agent connected to the DT, building the basis of the human–AI workflow. It will associate physical bridge components with current observations, historical changes, and sensor information that the user can query to build SA.

T3.2 Immersive Component (XR): T3.2. will enable immersive processes for experiencing the DT. Together with stakeholders, XR will be explored for their use with DTs for on-site and remote use, and paired with the workflow presented in T3.1.

T3.3 Experimental Evaluation: The framework will be evaluated in laboratory studies with realistic bridge assessment scenarios and in the field. Experienced participants will perform representative tasks (e.g., identification of deterioration mechanisms, planning of inspections) with conventional engineering workflows, with AI assistance without semantic grounding, and the proposed knowledge-grounded human–AI workflow.

Developed technology impact: Faster interpretation of inspection information, less effort searching fragmented records, better prioritization of inspection resources, and traceable evidence supporting engineering decisions.

2.4 Research Environment and Supervision

The PhD project is hosted at AU, with Assoc. Prof. Giuseppe Abbiati as main supervisor. The PhD Candidate (Lorenzo Loyola) will spend approximately 18 months at ETHZ.

Assoc. Prof. Giuseppe Abbiati (AU) has expertise in DT engineering for SHM and will support the implementation of the DTs for the bridge case-studies.

Assist. Prof. Jhonattan G. Martinez (AU) has expertise in automated inspections and will support the formulation of the semantic representation and its integration with LLMs.

Prof. Christos Georgakis (AU) has expertise in bridge engineering and SHM and will support the formulation of the inspection plans for the bridge case-studies.

Dr. Carolin Reichherzer (ETHZ) has expertise in human-computer interaction in immersive environments. She will support development of human-centred XR immersive environments.

Prof. Ueli Angst (ETHZ) has expertise in material durability and will supervise the formulation of the reasoning chain for deterioration assessment for the bridge case-studies.

2.5 Stakeholders involved in the project

Besides the project partners (AU and ETHZ), **HAI-BRI** involves the following stakeholders, both situated in the Global South:

- **Prof. Dayana Bastos, Federal University of Bahia School of Engineering (FUBSE), Department of Structural and Construction Engineering, Brazil:** structural engineering and SHM expertise; access to the Ponte do Funil.
- **Prof. Peter Njeru Njue, University of Nairobi (UN), Department of Real Estate, Construction Management and Quantity Surveying, Kenya:** geomatics and construction management expertise; access to the elevated viaducts of the Nairobi Expressway.

3 Project relevance to CEBE research fields

RF7 – Extending the life of the built environment (primary). **HAI-BRI** addresses the RF7 mission of “decision support that integrates structural health monitoring information” [20, p. 132]. It builds on the AI/robot-assisted operation of SHM systems [20, p. 148] by adding the layer that turns their observations into condition hypotheses linked to deterioration mechanisms such as rebar corrosion. It contributes to practical guidance for risk-informed decisions on repair, retrofitting, and transformation [20, p. 145] by quantifying how engineers rely on AI-supported assessments. It addresses the limited confidence in decisions based on scarce monitoring data and the adoption of new digital tools by practitioners [20, p. 141].

RF4 – Digitalization and automation. The semantic DT and KG of WP1 extend the semantic modeling, reality capture, and interoperability of WP4.4–4.6 [20, pp. 79–80] from buildings to in-service bridge infrastructure. The project will contribute to the RF4/RF7 knowledge exchange on DT engineering and agentic AI [20, p. 146].

RF5 – Climate resilient and adaptive infrastructure. Water ingress is driven by precipitation and exposure. The KG will represent environmental exposure as context for interpreting moisture and corrosion symptoms, interfacing with RF5 work on climate and infrastructure data (WP5.1)[20, p. 102].

RF1 – Assessing and measuring environmental sustainability. The environmental consequences of case-study decisions will be screened with the RF1 sustainability framework[21, p. 24].

4 Sustainability goals of the project and relation to the CEBE program vision

HAI-BRI contributes to CEBE vision of a built environment in harmony with planetary boundaries by addressing risk-informed life extension (Gap 7, RF7) based on timely and trustworthy

interpretation of deterioration. In bridges, much avoidable material use and CO₂ stems from late detection. Failing expansion joints and drainage let water and chlorides reach the reinforcement, and a local repair escalates into deck rehabilitation or replacement.

Objectives. (O1) *Least intervention:* **HAI-BRI** supports decisions (inspect, repair, replace) that select the minimum action consistent with safety. (O2) *Information sufficiency:* **HAI-BRI** identifies the minimum information needed for a trustworthy assessment, measurements and computation are added only where they change the decision.

Trade-offs and methodological choices. Digitalization is not sustainable per se. LLM/VLM inference consumes energy and resources. **HAI-BRI** therefore adopts a fit-for-purpose principle. Existing records are exploited before new data are acquired. Postponing action is sustainable only if deterioration is not underestimated. The separation of evidence, prior knowledge and inference, and engineer oversight are the safeguards against automation bias.

SDG perspective: Bridge infrastructures connect settlements and guaranty the delivery of goods and services. The rise in temperature and increase in rainfall negatively impact the corrosion mechanisms affecting common construction materials like concrete and steel. Therefore **HAI-BRI** SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action).

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Giuseppe Abbiati

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ORCID: 0000-0002-5048-8505, **Scopus ID:** 37064203300, **Scopus h-index:** 14, **Scopus citations:** 817, **Pure:** <https://pure.au.dk/portal/en/persons/abbiati@cae.au.dk>

Education

- Ph.D. in Structural Engineering, University of Trento, Italy. Doctoral thesis title: “Dynamic substructuring of complex hybrid systems based on time integration, model-order reduction and dynamic identification techniques” (April 7, 2014).
- M.Sc. in Civil Engineering, Polytechnic of Turin, Italy (December 10, 2009).
- B.Sc. in Civil and Environmental Engineering, Polytechnic of Turin, Italy (October 21, 2008).

Employments

- Associate Professor–Head of Section for Structural Engineering, Department of Engineering, Section of Civil and Architectural Engineering, University of Aarhus, Denmark (2023-now).
- Tenure-Track Assistant Professor, Department of Engineering, Section of Civil and Architectural Engineering, University of Aarhus, Denmark (2019-2023).
- Post-doctoral researcher, Chair of Structural Dynamics and Earthquake Engineering, Department of Civil, Environmental and Geomatic Engineering, ETH Zurich, Switzerland (2014-2018).

Fellowships

Recipient of Swiss Government Excellence Scholarship for a 1-year post-doctoral fellowship at the Department of Civil, Environmental and Geomatic Engineering of ETH Zurich, Switzerland, 2014, REF. 2014.0588.

Research achievements and impact

Giuseppe Abbiati’s main research contributions relate to hybrid testing and digital twins for civil engineering structures (buildings, bridges, wind parks). His cumulative portfolio of funded research and innovation projects sums up to 2.9 MEUR, of which +2 MEUR are with industrial partners; +700 kEUR currently ongoing (all related to digital twins for the built environment).

Supervision record

Current: 5 PhD students, 2 postdocs, 7 master’s students.

Total: 7 PhD students (2 already graduated), 4 postdocs, 45 master’s students.

Member of the assessment committee for 9 doctoral thesis.

Teaching experience Course responsible for 15 ECTS in the MSc (finite-element analysis, experimental mechanics, machine learning for civil engineering) and 5 ECTS in the BSc (probability and statistics) of Civil and Architectural Engineering, Aarhus University.

Selected research project

1. Enhancing Hybrid Digital Twinning for the Preservation of Built Heritage (HERITWIN), 2026-2029, Marie Curie Staff Exchange Network, European Commission 123 kEUR (1.117 MEUR total budget)
2. Laboratories as data factories for DT of floating wind turbines operational from Day 1 (LAB4TWIN) 2026-2029 CET Partnership Joint Call 2025, 334 kEUR (1.160 MEUR total budget)

3. GIS-based platform for seismic resilience of urban communities (GURU), REF. 4339-00005B 2025-2028 (36m) EUREKA – Innovation Beyond Borders - 2024-23234/NP/Resilience Call 2024 298 kEUR (1.82 MEUR total budget)
4. Cyber-Physical Sensing for Machine and Structures (CP-SENS), REF. 2081-00006B 2022-2027 (48m) Danish Innovation Fund 480 kEUR (2.55 MEUR total budget).

Selected publications

Five most important (last 15 years):

1. Scussolini, L., Abbiati, G., & Ceravolo, R. (2026). Identification of Volterra time-varying parameters in dynamical systems under random excitation. *Journal of Sound and Vibration*, 645, 120128, DOI:10.1016/j.jsv.2026.120128.
2. Pizarro, D., M. Kovarbašić, G. Abbiati, and B. Stojadinović, (2025). “Multi-Axial Sub-assembly Testing System for Hybrid Simulation with Large-Scale Structural Components.” *Earthquake Engineering & Structural Dynamics*, 54(8):2084-2105, DOI:10.1002/eqe.4351.
3. Stamenov, D., Abbiati, G., & Sauder, T. (2025). Numerical estimation of generalized frequency response functions from time series data using NARX. *Mechanical Systems and Signal Processing*, 239, 113278, DOI:10.1016/j.ymssp.2025.113278.
4. G. Abbiati, M. Broccardo, I. Abdallah, S. Marelli, F. Paolacci, (2021). Seismic Fragility Analysis based on Artificial Ground Motions and Surrogate Modeling of Validated Structural Simulators, *Earthquake Engineering and Structural Dynamics*, 50(9): 2314-2333, ISSN 00988847, DOI:10.1002/eqe.3448.
5. G. Abbiati, V. La Salandra, O.S. Bursi, L. Caracoglia (2018). A Composite Experimental Dynamic Substructuring Method Based on Partitioned Algorithms and Localized Lagrange Multipliers. *Mechanical Systems and Signal Processing*, 100:85-112, ISSN 08883270, DOI:10.1016/j.ymssp.2017.07.020.
6. Abbiati G., R. Ceravolo, C. Surace (2015). Time-dependent Estimators for On-Line Monitoring of Full-Scale Structures under Ambient Excitation, *Mechanical Systems and Signal Processing*, 60:166-181, ISSN 08883270, DOI:10.1016/j.ymssp.2014.10.018.

Five recent, project-relevant (last 5 years):

1. P. Talasila, D. Tcherniak, A. M.D. Jensen, S Mahato, J. V. Medom, M. D. Ulriksen, G. Abbiati, A. Schörghofer-Queiroz, P. G. Larsen, L. Damkilde, (2026). A Digital Twin Platform for Structural Health Monitoring, *Computer-Aided Civil and Infrastructure Engineering*, 50:100086, ISSN 1093-9687, DOI:10.1016/j.cacaie.2026.100086.
2. Hansen, S. T., Thule, C., Gomes, C., Lausdahl, K. G., Madsen, F. P., Abbiati, G., & Larsen, P. G. (2024). Co-simulation at different levels of expertise with Maestro2. *Journal of Systems and Software*, 209, 111905, DOI:10.1016/j.jss.2023.111905.
3. Oakes, B, C Gomes, G Abbiati, E Kamburjan, E E. Baş, and S Engelsgaard. “Towards Ontological Service-Driven Engineering of Digital Twins.” In 1st International Conference on Engineering Digital Twins (EDTconf 2024). Linz, Austria: to appear, 2024, DOI:10.1145/3652620.3688261.
4. Baş, E. E., Abbiati, G., Gonçalves Gomes, C. Â., Jassmann, U., & Larsen, P. G. (2024). Digitalization of Large-Scale Testing Facilities for the Wind Industry: DIGIT-BENCH Digital Twin. *Journal of Physics: Conference Series*, 2767(4), 042033, DOI:10.1088/1742-6596/2767/4/042033.
5. Abbiati, G., C. Gomes, M. Sandberg, Z. Kazemi, S. T. Hansen, and P. G. Larsen. “Modelling for Digital Twins.” In *The Engineering of Digital Twins*, edited by J. Fitzgerald, C. Gomes, and P. G. Larsen, 89–127. Cham: Springer International Publishing, 2024. DOI:10.1007/978-3-031-66719-0_5.

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PROFILE

Industrial engineer and PhD in Engineering Science (Civil Engineering) with 15+ years of research and consultancy experience across Latin America, the United States, Europe, and the Middle East. Research focuses on data-driven construction management and Construction 4.0, integrating Digital Twins, BIM, UAS, IoT sensing, simulation, Lean Construction, AI-enabled decision support, and CO₂/energy monitoring. Experienced in interdisciplinary and industry-collaborative research, research funding, technology demonstrators, postgraduate supervision, and translation of digital methods into construction planning, safety, productivity, and sustainability applications.

ACADEMIC & PROFESSIONAL APPOINTMENTS

- Assistant Professor (Tenure Track), Aarhus University, Denmark | Jan 2024–Present
- Postdoctoral Researcher, Technion – Israel Institute of Technology | Mar 2021–May 2023
- Senior Consultant, PROCASUR Corporation, Chile | Mar 2016–Jun 2020

EDUCATION & RESEARCH TRAINING

Qualification	Period	Institution	Country
Postdoctoral Research Fellow	2021–2023	Technion – Israel Institute of Technology	Israel
PhD, Engineering Sciences (Civil Engineering)	2016–2020	Pontifical Catholic University of Chile	Chile
Research Scholar, Construction Management	2018–2020	University of Florida	United States
MBA	2012–2013	National University	Colombia
Industrial Engineering	2002–2007	University of the Atlantic	Colombia

RESEARCH EXPERTISE

Digital Twins • Construction 4.0 • AI-enabled decision support • BIM • UAS/UAV remote sensing • IoT sensing • simulation and agent-based modelling • Lean Construction • production planning and control • construction safety • productivity • energy and CO₂ monitoring • sustainable construction

SELECTED RESEARCH FUNDING & PROJECT LEADERSHIP

- **Smart Infrastructure Emissions Intelligence (SIEI): A Decision-Support System for Sustainable Civil Infrastructure** Principal Investigator, Danish Business Authority (2026–2028)
- **GIS-based Platform Utilized for Seismic Resilience of Urban Communities (GURU)** Co-PI, EUREKA (2025/26–2028)
- **NextGen Build: Empowering Green Innovation and Carbon Intelligence in African and Danish Cities** — Principal Investigator, Danida (2026–2027)
- **Global Innovation Network Program (GINP)** Co-Investigator. Danish Agency for Higher Education and Science (2025–2026)
- **Predictive Carbon Emissions Monitoring and Control System for Earthwork and Interior Work** — Principal Investigator, We Build Denmark (2024–2025)
- **BIM2TWIN: Optimal Construction Management & Production Control** Senior Researcher, Horizon Europe (2020–2023)
- **Young Leaders Program for Rural Development in Central America** Senior Consultant, IFAD (2019–2020)
- **Advanced Monitoring of Construction Safety Using UAS and 3D Models**, Principal Investigator, Pontifical Catholic University of Chile (2018)
- **Strengthening Capacities and Tools to Disseminate Innovations**, Senior Consultant, IFAD (2016–2018)

SELECTED JOURNAL PUBLICATIONS

- Martinez, J. G., Murguía, D., Osorio, C., Gimenez, Z., & Yeung, T. Digital twin construction readiness: Development and validation of an organizational assessment instrument. *Journal of Construction Engineering and Management*. Under review.
- Loyola, L. I., & Martinez, J. G. (2025). Real-time energy and CO₂ optimization in construction cranes through digital twins and lean construction. *Revista Ingeniería de Construcción*, 40(Special Issue), 1–11. <https://doi.org/10.7764/RIC.00166.21>
- Yeung, T., Martinez, J. G., Schlenger, J., Borrmann, A., & Sacks, R. (2025). Integrating digital twin and agent-based simulation to support adaptive system design in building projects. *Automation in Construction*, 180, 106550. <https://doi.org/10.1016/j.autcon.2025.106550>
- Martinez, J., Yeung, T., Sacks, R., Shahaf, Y., & Sharoni, L. (2023). Situational awareness in construction uses a serious game. *Journal of Construction Engineering and Management*. <https://doi.org/10.1061/JCEMD4.COENG-12521>
- Martinez, J., Albeaino, G., Gheisari, M., Issa, R. R. A., & Alarcón, L. F. (2021). iSafeUAS: An unmanned aerial system for construction safety inspection. *Automation in Construction*, 125, 103595. <https://doi.org/10.1016/j.autcon.2021.103595>
- Martinez, J., Albeaino, G., Gheisari, M., Volkmann, W., & Alarcón, L. F. (2021). UAS point cloud accuracy assessment using SfM-based photogrammetry and PPK georeferencing technique for building surveying applications. *Journal of Computing in Civil Engineering*. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000936](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000936)
- Martinez, J., Gheisari, M., & Alarcón, L. F. (2020). UAV integration in current construction safety planning and monitoring processes. *Journal of Management in Engineering*, 36(3), 05020005. [https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000761](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000761)

BOOK / BOOK CHAPTER

- Martinez, J. G., Alarcón, L. F., & Wandahl, S. (2024). Unmanned Aerial Systems (UAS)-Derived 3D Models for Digital Twin Construction Applications. <https://doi.org/10.5772/intechopen.1004746>

RESEARCH IMPACT & ACADEMIC INDICATORS

- 7 journal publications with selective editorial policy; 15 conference publications; 2 book chapters; 1 book.
- 590+ Google Scholar citations (reported September 2026).
- Master's supervision: 10 completed dissertations and 3 ongoing.
- Technology translation: 2 software prototypes for construction planning, optimization, and CO₂ emissions; 3 optimized processes implemented.
- Industry engagement: technical/scientific consultancy with 22 organizations across three continents; 2 companies created/supported.

TEACHING, SUPERVISION & SERVICE

- Teaching: Construction 4.0, Aarhus University; topics include BIM/CDE, cyber-physical systems, Digital Twins, IoT, autonomous construction, robotics, and extended reality.
- Postgraduate supervision in digital construction, resilience, construction safety, sensing, Digital Twins, productivity, and sustainability.
- Reviewer for *Automation in Construction*, *Advanced Engineering Informatics*, and *Journal of Management in Engineering*.
- International research collaborations and networks spanning Europe, North America, Latin America, and the Middle East.

SELECTED AWARDS

- Best UAS design for safety applications, IDC 2020.
- Postdoctoral and doctoral scholarships supporting international research training.

ONLINE PROFILES

Aarhus University profile: [https://cae.au.dk/en/about-the-](https://cae.au.dk/en/about-the-department/contact/employees/show/person/jgmartinez%40cae.au.dk)

[department/contact/employees/show/person/jgmartinez%40cae.au.dk](https://cae.au.dk/en/about-the-department/contact/employees/show/person/jgmartinez%40cae.au.dk) | **ORCID:** <https://orcid.org/0000-0001-6715-4440> |

Google Scholar: <https://scholar.google.com/citations?user=iSNe3tQAAAAJ&hl=en>

CEBE Interdisciplinary Fellowship Programme 2026

Curriculum Vitae

1. Personal and Academic Information

Name: Christos T. Georgakis

Title: Professor

Affiliation: Aarhus University

Research group: Structural Dynamics and Geotechnical Engineering; CEBE RF7 – Extending the Life of the Built Environment

PhD year: 2001

ORCID: 0000-0001-6001-3856

2. Research Achievements and Relevant Impact

Research achievements:

Christos T. Georgakis is Professor of Structural Dynamics and Monitoring at Aarhus University and Lead of CEBE Research Field 7, *Extending the Life of the Built Environment*. His research spans structural health monitoring, bridge engineering, structural dynamics, wind engineering, vibration control and environmental loading, with particular emphasis on methods that enable safer and longer use of existing infrastructure. His work has combined experimental mechanics, full-scale monitoring, wind-tunnel testing, operational modal analysis, virtual sensing and, more recently, machine-learning-based prediction and nowcasting of extreme environmental events. He is co-author of *Cable Supported Bridges: Concept and Design* and has led or participated in numerous interdisciplinary and industry-funded research programmes.

Impact:

His research has been applied to major international infrastructure including the London Millennium Bridge, Queensferry Crossing, Port Mann Bridge and Gordie Howe International Bridge, and he led the independent investigation of the Chirajara Bridge collapse. His work has also generated patented vibration- and cable-control technologies, including solutions licensed to industry. In 2026 he received the **Robert H. Scanlan Medal** from ASCE's Engineering Mechanics Institute for extraordinary contributions to bridge, aerodynamics, and wind engineering.

Relevance to the proposed project:

Within CEBE RF7, Georgakis contributes senior expertise in structural health monitoring, predictive monitoring of extreme events, infrastructure dynamics and industry translation. His role is particularly relevant to projects that use sensing, modelling and prediction to support condition assessment, risk-informed intervention and safe life extension of existing structures.

3. Supervision and Research Training

<i>Category</i>	<i>Current</i>	<i>Former</i>
PhD students	3	19
Postdoctoral researchers	0	8
Master's students	1	80

4. Selected Publications

A. Five Most Important Publications — Last 15 Years

1. **Gimsing, N.J. & Georgakis, C.T.** *Cable Supported Bridges: Concept and Design*, 3rd ed. Wiley, 2012. DOI: 10.1002/9781119978237.
2. **Tarpø, M.G., Nabuco, B., Georgakis, C.T. & Brincker, R.** “Expansion of experimental mode shape from operational modal analysis and virtual sensing for fatigue analysis using the modal expansion method.” *International Journal of Fatigue*, 130, 105280, 2020. DOI: 10.1016/j.ijfatigue.2019.105280
3. **Demartino, C., Koss, H.H., Georgakis, C.T. & Ricciardelli, F.** “Effects of ice accretion on the aerodynamics of bridge cables.” *Journal of Wind Engineering and Industrial Aerodynamics*, 138, 98–119, 2015. DOI: 10.1016/j.jweia.2014.12.010.
4. **Chen, J. & Georgakis, C.T.** “Tuned rolling-ball dampers for vibration control in wind turbines.” *Journal of Sound and Vibration*, 332(21), 5271–5282, 2013. DOI: 10.1016/j.jsv.2013.05.019.
5. **Maienza, C., Avossa, A.M., Ricciardelli, F., Coiro, D., Troise, G. & Georgakis, C.T.** “A life cycle cost model for floating offshore wind farms.” *Applied Energy*, 266, 114716, 2020. DOI: 10.1016/j.apenergy.2020.114716.

B. Five Recent Publications Relevant to the Proposed Project — Last 5 Years

6. **Georgakis, C.T., Lagaros, N.D., Demartino, C. & Modé, P.D.** “Predictive Monitoring of Extreme Events for Enhanced Bridge Safety Using Hyperlocal ML-Powered Wind Nowcasting.” *e-Journal of Nondestructive Testing*, 31(2), 2026. DOI: 10.58286/32477. **Project relevance:** Predictive monitoring and short-term extreme-event decision support for infrastructure.
7. **Wei, J., Narazaki, Y., Quaranta, G., Yang, Q., Georgakis, C.T. & Demartino, C.** “Wind profile nowcasting and forecasting using machine learning.” *Journal of Wind Engineering and Industrial Aerodynamics*, 266, 106162, 2025. DOI: 10.1016/j.jweia.2025.106162. **Project relevance:** ML-based environmental sensing and prediction for structural and infrastructure applications.
8. **Modé, P., Demartino, C., Georgakis, C.T. & Lagaros, N.D.** “Short-term extreme wind speed forecasting using dual-output LSTM-based regression and classification model.” *Journal of Wind Engineering and Industrial Aerodynamics*, 259, 106035, 2025. DOI: 10.1016/j.jweia.2025.106035. **Project relevance:** Data-driven prediction of rare environmental hazards affecting infrastructure safety and operation.
9. **Kristoffersen, J.C., Kabel, T., Georgakis, C.T., Bellos, V. & Karpadakis, I.** “Spatio-temporal measurement of laboratory wave fields using LiDAR.” *Coastal Engineering*, 209, 105047, 2026. DOI: 10.1016/j.coastaleng.2026.105047. **Project relevance:** High-resolution, non-contact sensing and environmental load characterisation.
10. **Matejčka, L., Delpech, P., Georgakis, C.T. & Flamand, O.** “An experimental investigation of ice and snow shedding from bridge cables with and without surface modifications.” *Cold Regions Science and Technology*, 220, 104148, 2024. DOI: 10.1016/j.coldregions.2024.104148. **Project relevance:** Monitoring, understanding and mitigation of climate-related hazards affecting bridge operation and service life.

Relevant industry/public-sector partnerships: *International bridge engineering, monitoring and forensic assignments include the Gordie Howe International Bridge, Queensferry Crossing, Port Mann Bridge, Wittpenn Bridge and the independent investigation of the Chirajara Bridge collapse.*

Selected awards/recognition: *2026 Robert H. Scanlan Medal*, American Society of Civil Engineers, Engineering Mechanics Institute, *2009 Reinholdt W. Jørck and Family Foundation Career Achievement Award*.

Ueli M. Angst

Curriculum Vitae – September 2026

Personal details

First & FAMILY name:	Ueli Michael ANGST	Date of birth:	28 August 1980
Nationalities:	Swiss and French	Status:	Married, 3 children
Telephone:	+41 (0)79 944 03 40	E-mail:	uangst@ethz.ch
Researcher ID:	M-6348-2014	Google Scholar:	Link to profile
Website:	http://www.ifb.ethz.ch/durability		

Summary

Associate Professor at ETH Zurich and head of the Durability of Engineering Materials group, focusing on corrosion mechanisms, durability of infrastructure, electrochemistry, transport in porous materials, condition assessment, monitoring, and corrosion mitigation. My research combines experimental and computational methods from corrosion science, materials science, civil engineering, and reactive mass transport to improve the assessment, prediction, and service-life management of engineered materials and structures.

Key facts: 140+ scientific journal publications and 150+ conference contributions; 2 patent applications filed.

Current and previous positions

Since 06/2024	Associate professor at ETH Zurich , Switzerland, heading the research group Durability of Engineering Materials
Since 08/2021	Co-founder of ETH-SpinOff DuraMon
Since 04/2017	President of the Swiss Society for Corrosion Protection (SGK), which is the leading agency in corrosion consulting in Switzerland (and which is active worldwide)
Since 04/2012	Member of the board of directors at the material testing institute IMP Bautest AG, Switzerland
01/2017–05/2024	Assistant professor at ETH Zurich , Switzerland
08/2011 – 12/2016	Postdoctoral researcher at the Institute for Building Materials , ETH Zurich , Switzerland (part time, var. 40-60%), with Prof. Dr. R.J. Flatt and Prof. Dr. B. Elsener. <i>Research, teaching, and supervising in the field of durability of concrete structures.</i>
08/2011 – 11/2016	Consulting engineer at the Swiss Society for Corrosion Protection (SGK), Zurich, Switzerland (part time, 40%). <i>Consultant for corrosion in the construction industry and energy sector; various research projects.</i>

Education

04/2007 – 05/2011	PhD thesis at the Norwegian University of Science and Technology , NTNU, Trondheim, Norway (PhD defence: 03 May 2011). Supervisor: Prof. Ø. Vennesland Topic: Chloride-induced reinforcement corrosion in concrete
10/2000 – 01/2005	Studies in civil engineering at ETH Zurich , Switzerland (diploma equivalent to MSc degree)

Awards and recognitions (selection, for full list see [research group website](#))

ERC Starting Grant (2019); **RILEM Robert L'Hermite Medal** (2017); **SNSF Professorship Grant** (2016); 10+ Best/Outstanding paper awards (Guy Bengough, Le Chatelier, etc.); best doctoral dissertation award of the German Chemical Society GDCh (2012); **Dandelion Entrepreneurship Award (2021)** by ETH Entrepreneur Club & AI Center recognizing professors for their efforts to promote entrepreneurship.

Editorial work

Editor: *CEMENT* (Elsevier, since 2020); **Associate editor:** *RILEM Technical Letters* (since 2017) and *Materials & Structures* (since 2026); **Editorial board member:** *Corrosion Communications* (since 2023); Guest editor of special issues in *Transport in Porous Media*, *CEMENT*, and *Materials and Corrosion*.

Scientific society memberships and committee work

- [RILEM](#) member and deputy chair of TC [DLC](#) (since 2026), deputy chair of TC [296-ECS](#) (2020–2026) chair of TC [262-SCI](#) (2014–2021). Member of various RILEM TCs.
- Chair of the GfKORR [working group](#) “Korrosionsschutz im konstruktiven Ingenieurbau” (since 2019).
- Steering committee of the [Swiss Corrosion Science Network](#); member of [fib](#), [DGZfP](#), [ACI](#), and [AMPP](#).

Service to the academic community and supervision

Management, faculty service, supervision

- Since 2024, **Director of Studies** for the BSc, MSc, and PhD programs in civil engineering at ETH Zurich, responsible for teaching operations, quality management, and program development.
- Supervised 14 completed PhD theses, including 11 as main advisor; supervised 80+ BSc/MSc theses and served as examiner for 25+ PhD theses internationally.

Outreach and innovation

Co-founder of ETH spin-off [DuraMon](#) and host laboratory of ETH spin-off [Talpa Inspection](#); two European patent applications filed on corrosion assessment and sensor technologies.

Publications (selection)

Researcher ID: [M-6348-2014](#); Google Scholar: [Link to profile](#); Scopus: [ID 22978555600](#)

Five most important publications (from the last 15 years)

- Albert C, Schmid T, Zhang Z, **Angst U**. Rethinking concrete durability for low-carbon concretes through climate-informed corrosion modelling. *Nature Communications* (2026) [doi.org/10.1038/s41467-026-76916-3](#)
- Governo S, Rossi E, Azad S, Kaestner A, **Angst U**. The interdependent roles of chloride, moisture and dissolved oxygen in macro-voids at the steel-concrete interface in corrosion. *Corrosion Science* 265 (2026) 113814. [doi.org/10.1016/j.corsci.2026.113814](#)
- Albert C, Mundra S, Ferreira Sanchez D, Furcas F, Rajyaguru AD, Isgor B, Grolimund D, **Angst U**. Microscale chemical imaging to characterize and quantify corrosion processes at the metal-electrolyte interface. *npj Materials Degradation* 8 (2024) 116. [doi.org/10.1038/s41529-024-00534-x](#)
- Stefanoni M, **Angst U**, Elsener B. Kinetics of electrochemical dissolution of metals in porous media. *Nature Materials* (2019). [doi:10.1038/s41563-019-0439-8](#)
- **Angst U**, Elsener B. The size effect in corrosion greatly influences the predicted life span of concrete infrastructures. *Science Advances* 3 (2017) e170075. [doi.org/10.1126/sciadv.1700751](#)

Five recent for the project relevant publications (from the last 5 years)

- Furcas F, Vogel F, Lothenbach B, Angst U. Mechanism of crevice corrosion in historic riveted steel structures. *Communications Engineering* (2026). [doi.org/10.1038/s44172-026-00780-8](#)
- Yilmaz D, Thoma K, **Angst U**. A probabilistic framework for modelling chloride-induced corrosion and related load-deformation behaviour of reinforced concrete. *Structural Concrete* (2026). [doi.org/10.1002/suco.70776](#)
- **Angst U**, Rossi E, Boschmann Käthler C, Mannes D, Trtik P, Elsener B, Zhou Z, Strobl M. Chloride-induced corrosion of steel in concrete – insights from bimodal neutron and X-ray microtomography combined with ex-situ microscopy. *Materials and Structures* 57 (2024) 56. [doi.org/10.1617/s11527-024-02337-7](#)
- Pfändler P, Bodie K, Crotta G, Pantic M, Siegwart R, **Angst U**. Non-destructive corrosion inspection of reinforced concrete structures using an autonomous flying robot. *Automation in Construction* 158 (2024) 105241. [doi.org/10.1016/j.autcon.2023.105241](#)
- Pfändler P, Bircher L, **Angst U**. Inspecting the corrosion state of underground reinforced concrete structures. *Journal of Infrastructure Preservation and Resilience* 3 (2022) 17. [doi.org/10.1186/s43065-022-00064-3](#)

Carolyn Reichherzer

[email](#)

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[ORCID](#)

Switzerland

Dept. of Civil, Environmental and Geomatic Engineering | ETH Zurich

Research Profile: Across academic and industry roles, I have designed, developed, and evaluated how immersive technology shapes perception, decision-making, and user behaviour, with the intention of better understanding novel technologies and improving how we engage with them. I am now leveraging this expertise to investigate how XR combined with AI can shape and augment user trust and behaviour in human-AI interaction, with a particular focus on transparent and reliable decision support. With the rise of more accessible form factors in immersive technologies, better sensors, and the integration of AI, I aim to be a pioneer in developing new ways in which these technologies can be used responsibly and effectively within civil engineering.

Education

2021 - PhD in Computer Science | University of South Australia (now Adelaide University)

Master of Science in Digital Media | University of Bremen

Bachelor of Arts in Communication Design | University of Applied Sciences Mainz

Experience

Postdoctoral Fellow | ETH Zürich

July 2026 – Present

- Investigating the use of Augmented Reality visualisation techniques and LLM traits to enhance collaboration, trust and appropriate reliance for effective decision-making.
- Developing novel interaction paradigms for a new generation of lightweight XR headsets in combination with conversational agents.
- Co-Leading the Immersive Design Lab, providing XR expertise and resources to AEC researchers at ETH.

Senior Research Associate | Lucerne University of Applied Sciences and Arts

Nov 2024 – June 2026

- Led mixed-methods studies on photorealistic immersive experiences with novel 3D representation, evaluating perception and user impact to inform design for immersive experiences, and conducted statistical analyses.
- Initiated research on how LLM personality traits, context and embodiment influence user trust, with the goal of informing conversational AI design practices.
- Contributed to experimental design for a project on conversational agents for spiritual guidance.

UX Researcher | Meta

March 2023 – September 2024

- Evaluated the avatar motion quality of AI-driven tracking technology in Virtual Reality via lab experiments.
- Led foundational and generative research on the quality of photorealistic avatars.
- Under uncertainty, steered strategy and direction based on study outcomes, influenced product roadmaps, and communicated actionable research findings to various stakeholders clearly and concisely.
- Built and managed the first User Experience Research lab in Zurich to drive research initiatives in this location.

Senior Research Associate | Lucerne University of Applied Sciences and Arts

April 2021 – March 2023

- Facilitated evaluative and generative research on usability and perception for various AR/VR applications via lab experiments, surveys, interviews, and logged user behaviour. Conducted qualitative and quantitative analyses.
- Project lead of cross-disciplinary projects with external stakeholders to align on needs and priorities, and presented research insights to both academic and industry partners.

Research Associate | Australian Research Centre for Interactive and Virtual Environments

October 2020 – March 2021

- Led investigations and co-developed novel cross-device user experiences, specifically exploring the simultaneous interaction between head-mounted displays and mobile devices to assess how we can better work with AR.
- Evaluated diverse user interfaces through rigorous qualitative and quantitative user studies, assessing UX quality and the impact of critical factors like field of view on user performance and perception.

PhD Candidate | Australian Research Centre for Interactive and Virtual Environments

December 2016 – March 2021

- Effectively collaborated with a diverse group of law professionals and scientists via workshops and expert interviews to set the foundation for utilising VR for courtroom applications to improve decision-making and ease mental effort on trial participants.
- Developed prototypes in Unity and conducted user research using interviews, surveys, behavioural logging, eye-tracking, and physiological sensing to assess cognitive performance in immersive environments.

- Mentored students and facilitated undergraduate programming tutorials.

Research Assistant | University of South Australia (now Adelaide University)

October 2015 - April 2016

- Developed and evaluated hand-tracking interaction techniques for improved navigation of large datasets in VR using Leap Motion and Unity.

Research Assistant | Human Interface Technology Laboratory New Zealand

July 2013 – June 2014

- Developed an interactive prototype to explore creative solutions for sharing experiences remotely via graphical annotation with Google Glass.
- Planned, executed, and analysed user studies using subjective and objective measures for remote social connection with Google Glass.

Awards & Achievements

2022 **Best Paper Nomination (Top 20)**: IEEE VR 2022.

2021 **Honourable Mention – Best of CHI (Top 5%)**: CHI Conference 2021.

2020 **Research Excellence Awards Nominee**: University-wide nominee for STEM. University of South Australia.

2019 **1st Place Retail Game Changer Challenge**: \$25 000 AUD for winning prototype, Precision Group, Australia.

2019 **Highly Commended**: „Academic Excellence – Postgraduate Research“, South Australian government.

2019 **Shortlisted**: **Google Women Techmaker APAC** Finalist for the Asia-Pacific region (20 000+ applicants), Google.

Academic Service

since 2021 **Committee Member** for Conference on Human Factors in Computing Systems (CHI)

Reviewer for IEEE VR, ACM CHI, IEEE ISMAR, ACM TOCHI, PLOS, Journal of Human-Computer Interaction

Student Supervision

Master student (active): 1, Bachelor student (past): 1. Mentored 1 PhD student over 2 years with meetings 2x/month and 2 publications.

Community

2018 - 2021 **Women in STEM UniSA**: Co-founder and committee member, we won multiple awards for our engagement with students.

2017 - 2021 **Code Like a Girl**: Event volunteer and panellist, supporting the WSTEM community in Adelaide.

2019 **TechStars Startup Weekend Adelaide**: Tech Mentor.

Publications (most important)

[Honourable Mention] Reichherzer C. et al., Bringing the Jury to the Scene of the Crime: Memory and Decision-Making in a Simulated Crime Scene. In CHI Conference on Human Factors in Computing Systems (CHI '21), May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 12 pages, 2021.

[Best Paper Nomination] Reichherzer C. et al., „Supporting Jury Understanding of Expert Evidence in a Virtual Environment,“ 2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 2022, pp. 615-624, doi: 10.1109/VR51125.2022.00082.

[Selected for special issue journal publication] Reichherzer C. et. al, „Narrative and Spatial Memory for Jury Viewings in a Reconstructed Virtual Environment“ in IEEE Transactions on Visualization and Computer Graphics, 2018.

Reichherzer C. et. al., 2021. SecondSight: Demonstrating Cross-Device Augmented Reality. In ACM SIGGRAPH 2021 Immersive Pavilion (SIGGRAPH '21). Association for Computing Machinery, New York, NY, USA, Article 6, 1–2. <https://doi.org/10.1145/3450615.3464545>

Publications (most relevant)

Haslbauer, P., **Reichherzer, C.**, & Smolic, A. Deus in Machina: A Conversational AI for Spiritual Experience, International Conference on Digital and Interactive Arts, 2025.

[To Appear] Reichherzer C. et al., „User Perceptions of Gaussian Splats for Immersive Experiences,“ Quality and User Experience, Springer.

Reichherzer C. et al. 2021. SecondSight: A Framework for Cross-Device Augmented Reality Interfaces. In CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts), May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages, 2021.

Matthews, B. J., **Reichherzer, C.**, & Smith, R. T. (2023, April). Remapped interfaces: Building contextually adaptive user interfaces with haptic retargeting. In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (pp. 1-4).

Haslbauer, P., Pullen, M., Praprotnik, M., Stimec, S. P., Grosbois, A., **Reichherzer, C.**, & Smolic, A. (2025, June). VR Planica: Gaussian Splatting Workflows for Immersive Storytelling. In ACM International Conference on Interactive Media Experiences Workshops (IMXw) (pp. 47-53). SBC.

Lorenzo Loyola

Civil Engineer, MSc – PhD candidate (applicant)
Dept. of Civil and Architectural Engineering (CAE), Aarhus University, Denmark

ORCID: 0009-0005-2489-2677, **e-mail:** lorenzoloyola@cae.au.dk, **phone:** +45 71 39 37 98

Education

- M.Sc. in Civil and Architectural Engineering, Aarhus University, Denmark (2025). Thesis: “Digital Twin and Virtual Environment for Autonomous Bridge Inspection”, grade 12/12.
- Civil Engineer, Structural and Construction (professional degree), University of Chile (2018). Thesis: “Development of a BIM software application for interoperability between structural analysis and 3D reinforcement representation in reinforced-concrete walls”, grade 7.0/7.0.
- Bachelor of Engineering Sciences, University of Chile (2014).

Employments

- Research Assistant, Department of Civil and Architectural Engineering, Aarhus University, Denmark (2026-now): ontology and knowledge graph for structural monitoring data.
- Research Assistant (project-based), Aarhus University, Denmark (2025): real-time CO₂-monitoring digital twin for a construction site with Acembee, Clevertrack and NIRAS.
- Student Assistant, Aarhus University, Denmark (2024): national survey on Lean construction and health and safety.
- R&D Senior Consultant, Rene Lagos Engineers, Chile (2026-now, part-time, remote).
- Head of R&D, Rene Lagos Engineers, Chile (2021-2026): led a team of four engineers developing BIM automation tools.
- R&D Structural Engineer, Rene Lagos Engineers, Chile (2017-2021): reinforced-concrete design, structural instrumentation and monitoring, post-earthquake assessment.
- Assistant Engineer, Technical Office, Chile (2016-2017).

Research achievements and impact

Lorenzo Loyola’s research combines structural-engineering practice with digital construction. In industry he led the development of Revit add-ins that automate the transfer of structural design data into BIM reinforcement models, and carried out accelerometer-based monitoring and post-earthquake field assessments. At Aarhus University he deployed a real-time CO₂-monitoring digital twin on an active construction site with three industry partners (two first-author papers), built a BIM-based digital twin coupled to a Gazebo/ROS 2 virtual environment for simulated autonomous UAV bridge inspection (MSc thesis, submitted to i3CE 2026), and currently develops an ontology, Neo4j knowledge graph and LLM interface for structural monitoring data, the foundation of WP1 and WP2 of this proposal. Training: Urban Digital Twin PhD Spring School, London (2026); NRIS course on distributed LLM fine-tuning on HPC (2026).

Supervision record

Led a team of four engineers in industry (2021-2026).

Technical skills

Structural analysis (ETABS, SAFE, Robot, OpenSees, MATLAB); BIM automation (Revit API/C#, Dynamo, Rhino/Grasshopper, IFC); data and AI (Python, SQL, Neo4j, LLM APIs, LangChain, scikit-learn); simulation and robotics (ROS 2, Gazebo, ArduPilot, MAVLink, MAVSDK). Languages: Spanish (native), English (full professional), Danish (A2).

Selected publications

Five most relevant to the project:

1. Loyola, L., Martinez, J. G., & Abbiati, G. (2026). BIM-based digital twin for UAV bridge inspection. ASCE International Conference on Computing in Civil Engineering (i3CE 2026), submitted.
2. Loyola, L., et al. (in preparation). An ontology for structural monitoring data. Journal article.
3. Singh, S., et al., incl. Loyola, L. (2026). LLM-assisted chatbot for data-driven construction monitoring and decision support. AUC-CSCE Special Issue 50.
4. Loyola, L. I., & Martinez, J. G. (2025). Real-time energy and CO₂ optimization in construction cranes through digital twins and lean construction. Revista Ingeniería de Construcción, [10.7764/RIC.00166.21](#).
5. Loyola, L. (2025). Digital Twin and Virtual Environment for Autonomous Bridge Inspection. MSc thesis, Aarhus University.

Other recent publications (last 5 years):

1. Martinez Ribon, J. G., Montanari, P., Loyola, L. I., & Singhal, S. (2025). Uso de gemelos digitales y sensores IoT para la eficiencia energética de grúas en construcción. Simpósio Brasileiro de Tecnologia da Informação e Comunicação na Construção (SBTIC 2025), [10.46421/sbtic.v5i00.8211](#).
2. Ullah, A. N. L., Puja, T. C., Martinez, J., Loyola, L., & Perez, F. (2026). Proactive safety management in earthwork operations using IoT-driven real-time risk analysis. European Conference on Computing in Construction (EC3 2026).
3. Anam, et al., incl. Loyola, L. (2026). Requests for information in construction projects: comparing systematic literature evidence with industry survey findings. CIB W78 2026, New Delhi, India.
4. Pérez, C. T., Madushanka, M., Loyola, L., Ergul, M., & Salling, S. T. (2025). Health and safety perceptions of construction workers in Denmark. 23rd CIB World Building Congress, West Lafayette, USA, [10.7771/3067-4883.1125](#).
5. Pérez, C. T., Madushanka, M., Loyola, L., Ergul, M., Salling, S. T., & Wandahl, S. (2024). The impact of Lean knowledge and Lean operation on construction workers' job satisfaction. 32nd Annual Conference of the International Group for Lean Construction (IGLC32), [10.24928/2024/0188](#).



UNIVERSITY OF NAIROBI
DEPARTMENT OF REAL ESTATE, CONSTRUCTION MANAGEMENT AND
QUANTITY SURVEYING

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Tel: +254 020 3563018 Fax: + 254 02 318262
E-mail: recmq@uonbi.ac.ke

Ref.: REC/AU/CEBE/1/26
Date: 24th September 2026

Your Ref.:

Dear Sir/Madam,

RE: LETTER OF SUPPORT CEBE PhD PROJECT

Dear Members of the Selection Committee,

I am pleased to express my support for the PhD project "Human-AI Collaborative Workflows for Immersive Digital Twins in Bridge Monitoring and Inspection," led by Aarhus University together with its international research collaborators.

The proposed research addresses an important challenge for the management of critical infrastructure: transforming heterogeneous inspection, monitoring, visual, structural, and historical information into reliable and traceable evidence that can support engineering decisions. The integration of Digital Twins, Structural Health Monitoring, knowledge graphs, multimodal AI, and human engineering expertise is particularly relevant to infrastructure owners operating under different technological and resource conditions.

I am pleased to support the project through collaboration with the research team and the PhD candidate. In particular, I intend to contribute knowledge of infrastructure engineering and the local context in Kenya and to support the development of critical-infrastructure case studies for the research.

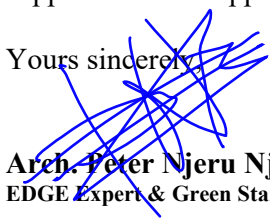
Subject to the relevant institutional and infrastructure-owner approvals, permissions, safety requirements, and data-access arrangements, I will assist the project team in identifying and facilitating access to suitable critical infrastructure in Kenya, including potential bridge case studies. Such cases may provide access to available inspection records, structural information, imagery, environmental observations, field measurements, and other relevant data that can contribute to the development and validation of the proposed Digital Twin and human-AI decision-support framework.

I am also willing to host the PhD candidate during a research stay at the University of Nairobi and facilitate engagement with relevant academic, professional, and infrastructure-management stakeholders. This research stay would support field data collection, case-study development, evaluation activities, and investigation of how the proposed framework performs under local infrastructure-management conditions.

The Kenyan case studies would provide an important international dimension to the project by enabling the researchers to examine the transferability of the proposed approach to environments where infrastructure data, monitoring technologies, historical records, and inspection resources may differ from those available in highly digitalized European contexts. This will strengthen the project's ability to develop methods that are not only technologically advanced but also adaptable to different infrastructure-management environments.

I look forward to collaborating with Aarhus University and the project partners and fully support this PhD application.

Yours sincerely,



Arch. Peter Njeru Njue

EDGE Expert & Green Star AP [MPhil, Cambridge, UK | B. Arch, UoN | MAAK(EDC) | ICPMK | ISOCARP]

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**FEDERAL UNIVERSITY OF BAHIA
SCHOOL OF ENGINEERING
DEPARTMENT OF STRUCTURAL AND
CONSTRUCTION ENGINEERING**



**LETTER OF SUPPORT
CEBE PhD Project**

September 23, 2026

To the CEBE Selection Committee

Dear Members of the Selection Committee,

I am pleased to express my support for the PhD project "Human-AI Collaborative Workflows for Immersive Digital Twins in Bridge Monitoring and Inspection," led by Aarhus University in collaboration with international research partners.

The project addresses an important challenge in the management of critical infrastructure by integrating Structural Health Monitoring, Digital Twins, semantic knowledge representation, multimodal artificial intelligence, and human-AI collaboration to support more informed and traceable infrastructure inspection and maintenance decisions.

I am pleased to support this research through collaboration with the project team and the PhD candidate. In particular, I intend to contribute expertise in structural engineering, infrastructure monitoring, and construction digitalization, and support the development of international case studies to evaluate the proposed methodology under real-world infrastructure conditions.

Subject to the necessary institutional approvals, permissions, safety requirements, and data-access agreements, I will support the identification of and facilitate access to suitable bridge and/or critical-infrastructure case studies in Brazil. These case studies can provide relevant inspection information, structural and environmental data, imagery, historical documentation, and other available evidence required for the development and validation of the proposed Digital Twin and AI-supported decision framework.

I am also willing to host the PhD candidate for a research stay in Brazil, providing an academic environment for collaboration with researchers and relevant infrastructure stakeholders. During this period, I will support the candidate in case-study definition, field activities, interpretation of local infrastructure conditions, and evaluation of the transferability of the developed methodology.

The Brazilian case studies are particularly relevant because they will enable the project to investigate the framework's applicability under infrastructure-management conditions that may differ substantially from those encountered in Denmark and Switzerland, including differences in climate, monitoring availability, historical information, and inspection and maintenance resources.

I look forward to collaborating with Aarhus University and the other project partners and fully support this PhD application.

Sincerely,

A handwritten signature in blue ink that reads "Dayana Bastos Costa". The signature is fluid and cursive, with the first name "Dayana" being the most prominent.

Professor Dayana Bastos Costa
Federal University of Bahia (UFBA)
Salvador, Bahia, Brazil

Budget

Item (DKK)	2027	2028	2029	Total
PhD salary	576,000	600,000	624,000	1,800,000
PhD school fees	80,000	80,000	80,000	240,000
Consumables: XR equipment	19,000	–	–	19,000
Consumables: cloud computing	15,000	15,000	15,000	45,000
Consumables: travel	12,000	12,000	12,000	36,000
Project supplement	250,000	250,000	250,000	750,000
Total requested from CEBE				2,890,000

The PhD candidate, Lorenzo Loyola, is already employed at Aarhus University and can start on 1 January 2027. The PhD salary is computed using the Aarhus University template for budgeting.

Consumables (equipment, running expenses, and travel) are kept within the DKK 100,000 limit of the call; XR equipment covers one lightweight VR or AR headset with accessories, depending on the on-site versus remote use case selected with the stakeholders. Costs for cloud computing are associated with AI developments. The travel budget will cover short staff exchanges.

The research stay at ETH Zurich will be funded through external mobility grants, for which the candidate will apply in the first project year. Danish foundations routinely support PhD stays abroad (e.g., [Otto Mønsted Fond, Augustinus Fonden, William Demant Fonden]), in addition to the mobility support of the AU Graduate School of Technical Sciences.